Step 1:

Identified Assets:

1. Device
2. Server
3. Communication Channel (Device <-> Server)

Asset 1: Device

STRIDE Category	Threat	Mitigation
Spoofing	Attackers impersonate the server to send malicious commands	Mutual authentication (e.g., mTLS with certificates)
Tampering	Firmware or configuration modified (e.g., malicious update)	Secure boot and signed firmware updates
Repudiation	Device denies sending certain data or actions	Signed logs
Information Disclosure	Sensitive data exposed from device storage	Encrypt data
Denial of Service	Device overwhelmed by excessive requests	Request validation
Elevation of Privilege	Exploiting software bug to gain admin access	Secure coding and sandboxing

Asset 2: Server

STRIDE Category	Threat	Mitigation
Spoofing	Fake device sends data pretending to be legitimate	Device authentication (certificates, device identity management)
Tampering	Data or commands modified on server	Integrity checks
Repudiation	N/A	N/A

STRIDE Category	Threat	Mitigation
Information Disclosure	Data breach exposing collected device data	Encryption and strict access control
Denial of Service	N/A	N/A
Elevation of Privilege	User escalates to admin role	Secure coding and sandboxing

Asset 3: Communication Channel

STRIDE Category	Threat	Mitigation
Spoofing	Man-in-the-middle (MITM) impersonates server/device	mTLS with certificate validation
Tampering	Data altered in transit	TLS integrity
Repudiation	Dispute over what was transmitted	Signed messages
Information Disclosure	Data intercepted over network	End-to-end encryption (TLS 1.2+)
Denial of Service	Network flooding	Rate limiting and firewall rules
Elevation of Privilege	Exploiting protocol weaknesses to inject commands	Strong protocol design and input validation

Step 2:

Function / Role	Device Agent	Operations User	Administrator
Collect data	✅	❌	❌
Send collected data	✅	❌	❌
Receive/view collected data	❌	✅	✅
Change data collection configuration	❌	❌	✅
Update device software	❌	❌	✅
Disable device	❌	❌	✅